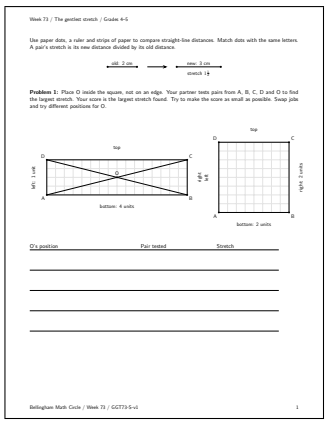
Week 73 / The gentlest stretch / Page menu

For adults. Band slices print the pages whose range includes the band, with page and problem numbers unchanged.
K–1: no pages · Grades 2–3: no pages · Grades 4–5: whole packet


Week 73 / The gentlest stretch / Grades 4–5

Use paper dots, a ruler and strips of paper to compare straight-line distances. Match dots with the same letters. A pair's stretch is its new distance divided by its old distance.

old: 2 cm → new: 3 cm — stretch 1½

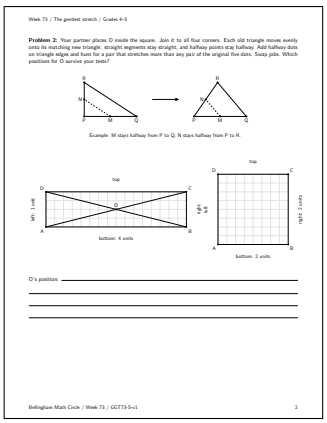
Problem 1: Place O inside the square, not on an edge. Your partner tests pairs from A, B, C, D and O to find the largest stretch. Your score is the largest stretch found. Try to make the score as small as possible. Swap jobs and try different positions for O.

top
D C
O
left: 1 unit
right: 2 units
A B
bottom: 4 units
bottom: 2 units

O's position	Pair tested	Stretch

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Problem 2: Your partner places O inside the square. Join it to all four corners. Each old triangle moves evenly onto its matching new triangle: straight segments stay straight, and halfway points stay halfway. Add halfway dots on triangle edges and hunt for a pair that stretches more than any pair of the original five dots. Swap jobs. Which positions for O survive your tests?

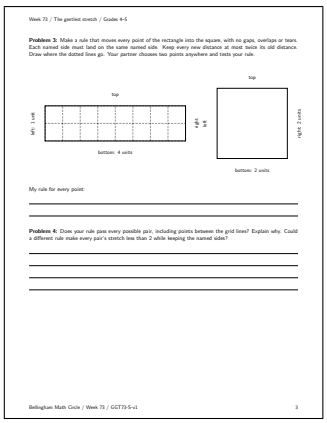
R
N
P M Q
Example: M stays halfway from P to Q. N stays halfway from P to R.

top
D C
O
left: 1 unit
right: 2 units
A B
bottom: 4 units
bottom: 2 units

O's position:

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Problem 3: Make a rule that moves every point of the rectangle into the square, with no gaps, overlaps or tears. Each named side must land on the same named side. Keep every new distance at most twice its old distance. Draw where the dotted lines go. Your partner chooses two points anywhere and tests your rule.

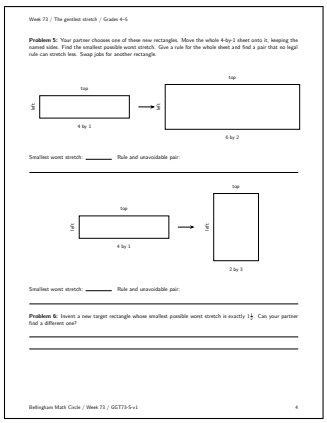
top
left: 1 unit
bottom: 4 units
top
left: 2 units
bottom: 2 units

My rule for every point:

Problem 4: Does your rule pass every possible pair, including points between the grid lines? Explain why. Could a different rule make every pair's stretch less than 2 while keeping the named sides?

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Problem 5: Your partner chooses one of these new rectangles. Move the whole 4-by-1 sheet onto it, keeping the named sides. Find the smallest possible worst stretch. Give a rule for the whole sheet and find a pair that no legal rule can stretch less. Swap jobs for another rectangle.

top
left
4 by 1
top
left
6 by 2

Smallest worst stretch: _____ Rule and unavoidable pair:

top
left
4 by 1
top
left
2 by 3

Smallest worst stretch: _____ Rule and unavoidable pair:

Problem 6: Invent a new target rectangle whose smallest possible worst stretch is exactly 1½. Can your partner find a different one?

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page 4 · 4–5 · P5–6